

St George Girls High School

Trial Higher School Certificate Examination

2011



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Begin each question on a new booklet
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Total Marks – 120

- Attempt ALL questions.
- All questions are of equal value.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks)	Marks
a) Simplify $\sqrt{75} - 2\sqrt{12}$	2
b) Solve $5x^2 = 2x + 3$	2
c) Factorise $y^3 - 8$	1
d) If $0^\circ \leq \theta \leq 360^\circ$ and $\cos\theta = -\frac{\sqrt{3}}{2}$ find all possible values of θ	2
e) If $y = xe^x$ find $\frac{dy}{dx}$	2
f) Convert 225° to radians .	1
g) Show on a neat diagram (at least $\frac{1}{3}$ of a page) the region defined by the intersection of: $x - y \leq -1$ and $5x + 3y \leq 19$	2

Question 2 – (12 marks)

Marks

- a) (i) Find the equations of the tangents to $y = 4x - x^2$ at the points where the curve crosses the x axis.

4

- (ii) Calculate the area bounded by the tangents and the x axis.

2

- b) Find the sum of the series

2

$$1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$$

- c) If $f(x) = x(x - 1)(x + 2)$ find the area bounded by $f(x)$ and the x axis between $x = -1$ and $x = 1$.

3

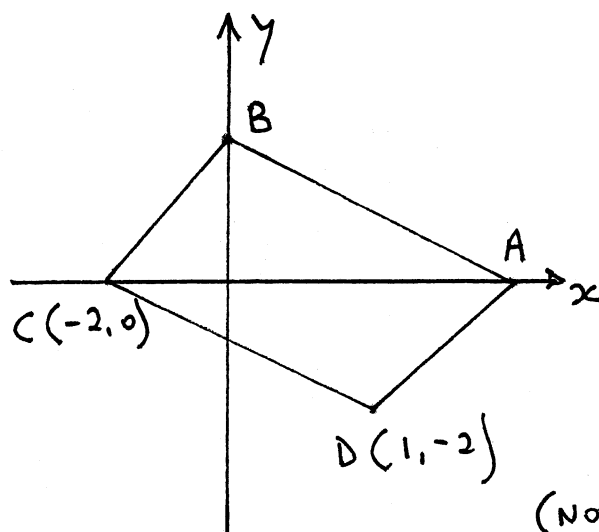
- d) Write as a logarithm $2^x = 7$

1

Question 3 – (12 marks)

Marks

a)



$ABCD$ is a quadrilateral as shown in the diagram. The equation of AB is $2x + 3y - 12 = 0$. The coordinates of C and D are $(-2, 0)$ and $(1, -2)$ respectively.

- (i) What is the size (to the nearest minute) of the acute angle between AB and the x axis.

2

- (ii) Show $AB \parallel CD$

1

- (iii) Show the equation of CD is $2x + 3y + 4 = 0$

2

- (iv) If M is the midpoint of AB , find the distance from M to CD

2

- (v) Hence, or otherwise, write down the equation of the circle centre M , having CD as a tangent.

1

b) Find:

(i) $\int \frac{3x}{x^2-1} dx$

2

(ii) $\int_{\pi}^{2\pi} \cos \frac{x}{3} dx$

2

Question 4 – (12 marks)

Marks

- a) Express $\frac{\sqrt{3}-1}{2\sqrt{3}-1}$ with a rational denominator. 2
- b) If $\frac{ds^2}{dt^2} = 3t - 4$ find s , given that when $t = 0$, $s = 6$ and $\frac{ds}{dt} = 0$ 2
- c) In $\triangle ABC$ $\sin B = \frac{2}{3}$, $a = 4$ and $b = 7$. Find $\sin A$ as a simple fraction. 2
- d) Find $\int \frac{x^2+x-1}{x^2} dx$ 2
- e) Draw a neat sketch of $y = 3 \sin 2x$ for $0^\circ \leq x \leq 360^\circ$ 2
- f) Solve $3 \times 9^x - 28 \times 3^x + 9 = 0$ 2

Question 5 – (12 marks)

Marks

- a) Differentiate with respect to x : $y = \frac{\tan x}{x}$ 2
- b) For the curve $y = x^3 - 3x^2$
- (i) Find the coordinates of any stationary points and determine their nature. 4
- (ii) Find the coordinates of any points of inflexion. 1
- (iii) Sketch the graph of this function, indicating clearly all relevant features. 3
- c) Find $\int \sqrt[3]{x} + \sin 3x \, dx$ 2

Question 6 - (12 marks)

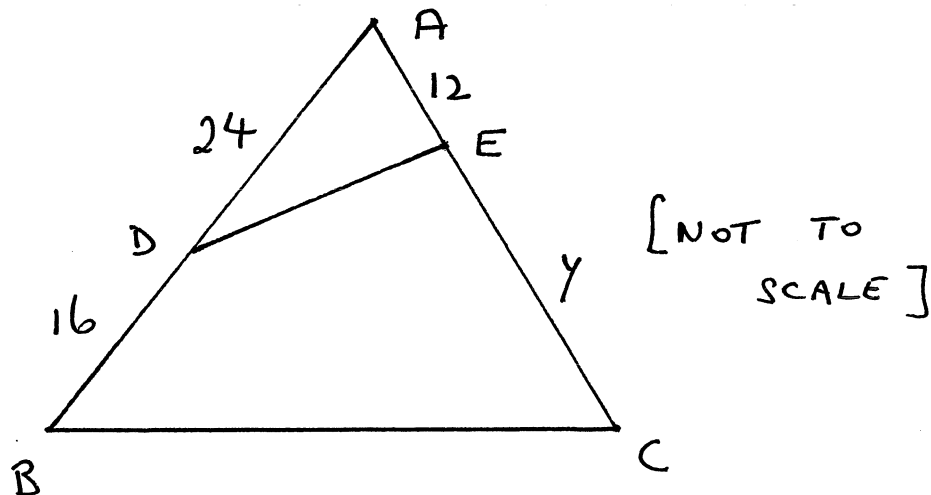
Marks

- a) Prove that

3

$$\frac{\sin^2 x}{1 - \cos x} + \frac{\sin^2 x}{1 + \cos x} = 2$$

- b)



In the diagram $\angle ADE = \angle ACB$. Prove that $\triangle AED \parallel \triangle ABC$

3

Hence find the value of y .

- c) The equation of a parabola is given by $x^2 - 4x - 2y + 8 = 0$. Find the:

(i) coordinates of the vertex.

2

(ii) coordinates of the focus.

2

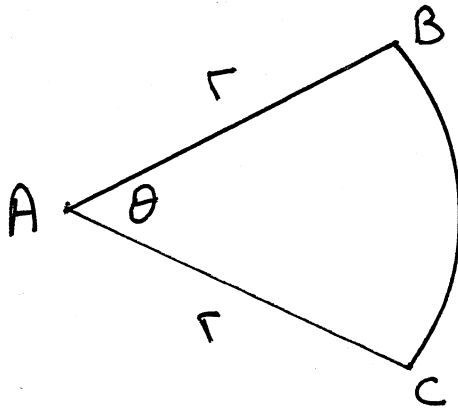
(iii) gradient of the normal to this parabola at the point $(0, 4)$

2

Question 7 – (12 marks)

Marks

a)



In the diagram AB and AC are radii of length r of a circle centre A . The arc BC subtends an angle of θ radians at A .

(i) Write down a formula for:

(α) the length of arc BC .

1

(β) the area of sector ABC .

1

(ii) The perimeter of ABC is 12m. Show that the area, y square metres, is given by

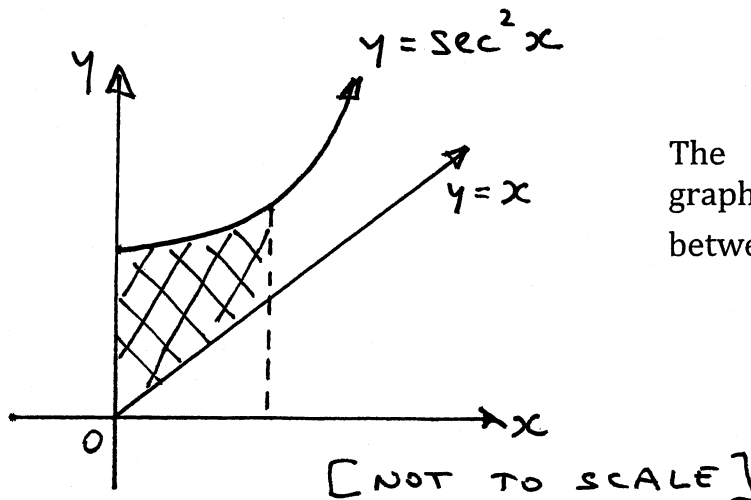
2

$$y = \frac{72\theta}{(\theta + 2)^2}$$

(iii) Show the maximum area is 9m^2 .

3

b)



The diagram shows the graphs of $y = \sec^2 x$ and $y = x$ between $x = 0$ and $x = \frac{\pi}{4}$

Calculate the area of shaded region, correct to 2 decimal places.

3

c) Find the values of m for which the equation $4x^2 - mx + 9 = 0$ has real roots.

2

Question 8 – (12 marks)

Marks

- a) The second term of a geometric series is $\frac{3}{8}$, and the seventh term is 12.
Find the 14th term.

2

- b) Let α and β be the roots of $3x^2 + 4x - 3 = 0$ evaluate

(i) $\alpha + \beta$

1

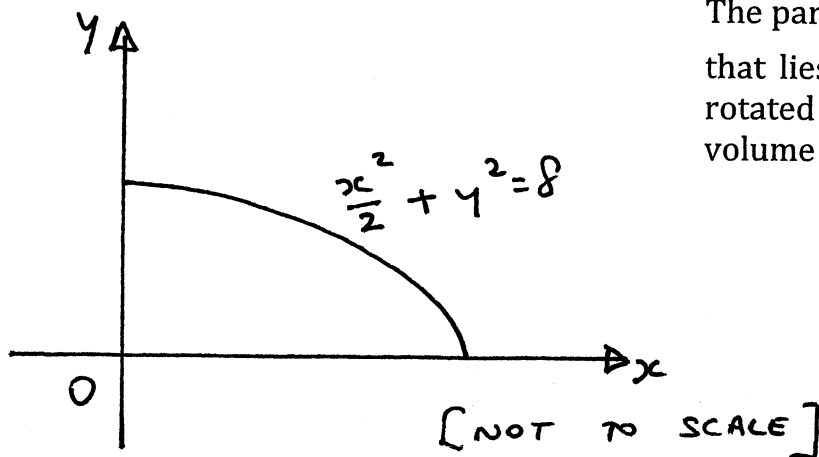
(ii) $\alpha\beta$

1

(iii) $2\alpha^2 + 2\beta^2$

2

c)



The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is rotated about the x axis. Find the volume of revolution.

3

- d) Grace invests \$10 000 at 6% p.a., interest calculated and added monthly, in a savings account.

- (i) What is the value of her investment after 2 years?

1

- (ii) What rate of simple interest would produce the same value over the 2 years (answer to 2 decimal places).

2

Question 9 – (12 marks)

Marks

a) (i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$

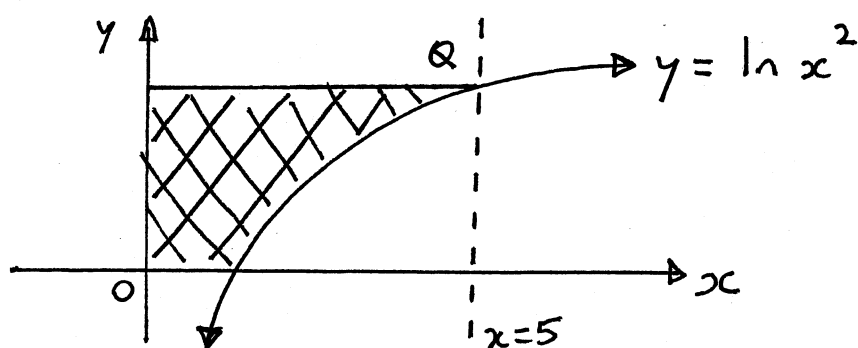
1

(ii) Using the relationship $\ln x^2 = 2 \ln x$, or otherwise, find

1

$$\int \ln x^2 dx$$

(iii)



3

The graph shows the curve $y = \ln x^2$ which meets the line $x = 5$ at Q . Using your answers from (i) and (ii), or otherwise, find the area of the shaded region.

b) (i) Write down the domain of

1

$$y = \frac{1}{x} + \log x$$

(ii) Show that the first and second derivatives may be expressed as

2

$$\frac{dy}{dx} = \frac{x-1}{x^2} \quad \text{and} \quad \frac{d^2y}{dx^2} = \frac{2-x}{x^3}$$

(iii) Show that the curve has a minimum at $(1, 1)$

2

c) Find k if

2

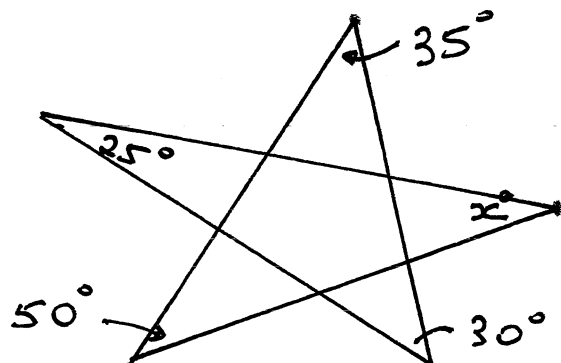
$$\int_1^k \frac{4}{x^2} dx = 3$$

Question 10 – (12 marks)

Marks

- a) Find the size of x , no reason required.

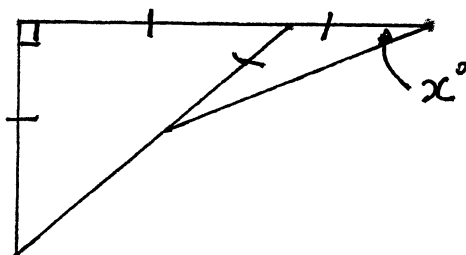
(i)



[NOT TO SCALE]

1

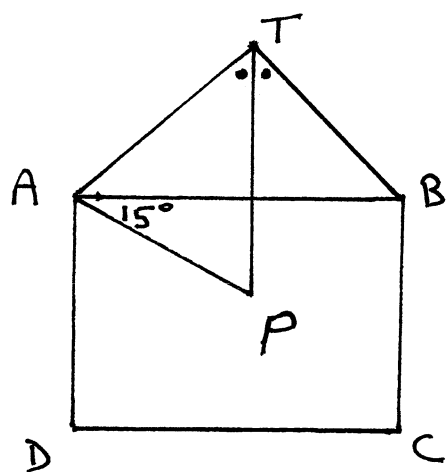
(ii)



[NOT TO SCALE]

1

b)



In the diagram $ABCD$ is a square and ABT is an equilateral triangle. The line TP bisects $\angle ATB$ and $\angle PAB = 15^\circ$

[NOT TO SCALE]

- (i) Copy the diagram into your answer booklet and show why $\angle PAT = 75^\circ$

1

- (ii) Prove $\triangle TAP \equiv \triangle DAP$

3

Question 10 – (cont'd)

Marks

- c) A car company offers a loan of \$20 000 to purchase a new car for which it charges interest at 1% per month. As a special deal the company does not charge interest for the first 6 months, however, the monthly payments start at the end of the first month. Beth takes out a loan and agrees to repay the loan over 5 years by making 60 equal monthly repayments of \$ m . Let A_n be the amount owing at the end of the n^{th} month (after a repayment and interest, if any).

- | | |
|--|---|
| (i) Find an expression for A_4 | 1 |
| (ii) Show $A_8 = (20\,000 - 6m)(1.01)^2 - m(1 + 1.01)$ | 1 |
| (iii) Find an expression for A_{60} | 1 |
| (iv) Find the value of m | 3 |

St George GHS Trial HSC Mathematics 2011

Question 1

(a) $5\sqrt{3} - 2 \times 2\sqrt{3}$
 $= \sqrt{3}$

(b) $5x^2 - 2x - 3 = 0$
 $(5x+3)(x-1) = 0$
 $x = -\frac{3}{5}, 1$

(c) $(y-2)(y^2+2y+4)$

(d) $\theta_{\text{acute}} = 30^\circ$
 $\theta = 180^\circ - 30^\circ, 180^\circ + 30^\circ$
 $= 150^\circ, 210^\circ$

(e) $y = xe^x$
 $\frac{dy}{dx} = 1 \cdot e^x + x \cdot e^x$
 $= e^x(1+x)$

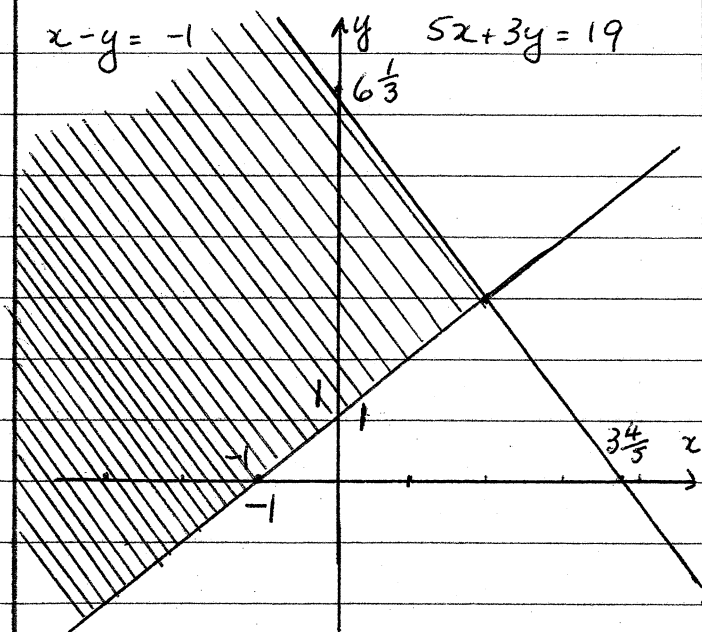
(f) $225^\circ = \frac{5\pi}{4} \text{ rad.}$

(g)

x	0	-1
y	1	0

x	0	$3\frac{4}{5}$
y	$6\frac{1}{3}$	0

$x-y = -1$ $5x+3y = 19$



Question 2

(a)(i) $y = 4x - x^2$

Cuts x axis when $y=0$

$4x - x^2 = 0$

$x(4-x) = 0$

$x = 0, 4$

$\frac{dy}{dx} = 4 - 2x$

When $x=0$ $\frac{dy}{dx} = 4$

$x=4$ $\frac{dy}{dx} = 4 - 2 \times 4 = -4$

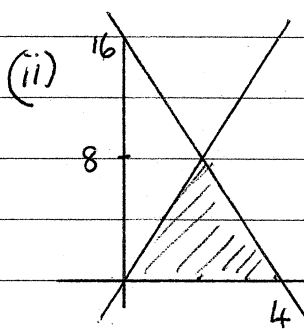
Eqⁿ of tangent at (0,0)

$y = 4x$

Eqⁿ of tangent at (0,4)

$y - 0 = -4(x - 4)$

$y = -4x + 16$



(ii)

$y = 4x$ mee

$y = -4x + 16$

$4x = -4x + 16$

$8x = 16$

$x = 2$

$y = 8$

$A = \frac{1}{2} \times 4 \times 8$

Area = 16 units²

(b) $1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$

Infinite geometric series

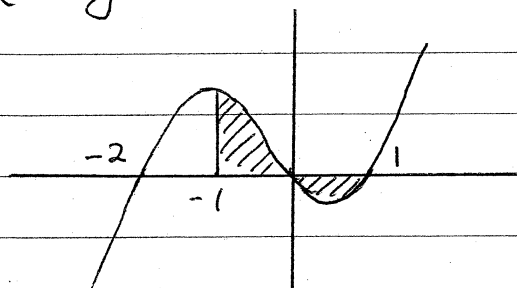
$a=1$ $r = -\frac{1}{4}$

$S = \frac{a}{1-r}$

$= \frac{1}{1 - (-\frac{1}{4})}$

$= \frac{4}{5}$

$$(c) y = x(x-1)(x+2)$$



$$x(x-1)(x+2) = x(x^2+x-2) \\ = x^3+x^2-2x$$

$$A = \int_{-1}^0 x^3+x^2-2x dx + \left| \int_0^1 x^3+x^2-2x dx \right| \\ = \left[\frac{x^4}{4} + \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[\frac{x^4}{4} + \frac{x^3}{3} - x^2 \right]_0^1 \right|$$

$$= 0 - \left(\frac{1}{4} - \frac{1}{3} - 1 \right) + \left| \left(\frac{1}{4} + \frac{1}{3} - 1 \right) - 0 \right|$$

$$= 1\frac{1}{12} + \left| -\frac{5}{12} \right|$$

$$= 1\frac{1}{2}$$

$$\text{Area} = 1.5 \text{ units}^2$$

$$(d) 2^x = 7$$

$$x = \log_2 7$$

Question 3

$$(a) (i) AB: 2x+3y-12=0$$

$$3y = -2x+12$$

$$y = -\frac{2}{3}x+4$$

$$\text{Grad of } AB = -\frac{2}{3}$$

Let θ be acute angle between

AB and the x axis

$$\tan \theta = \left| -\frac{2}{3} \right| = \frac{2}{3}$$

$$\theta = 33^\circ 41'$$

$$(ii) \text{Grad } CD = \frac{-2-0}{1--2}$$

$$= -\frac{2}{3}$$

$$= \text{Grad } AB$$

$$\therefore AB \parallel CD$$

$$(ii) \text{Eq}^n \text{ of } CD: m = -\frac{2}{3} \quad C(-2,0)$$

$$y-0 = -\frac{2}{3}(x--2)$$

$$3y = -2x-4$$

$$2x+3y+4=0$$

$$(iv) 2x+3y-12=0$$

$$\text{When } y=0 \quad x=6 \quad A(6,0)$$

$$x=0 \quad y=4 \quad B(0,4)$$

$$M \text{ is } \left(\frac{6+0}{2}, \frac{0+4}{2} \right) = (3, 2)$$

$$d = \frac{|2 \times 3 + 3 \times 2 + 4|}{\sqrt{2^2+3^2}}$$

$$= \frac{16}{\sqrt{13}}$$

$$(v) \text{Eq}^n \text{ of circle is}$$

$$(x-3)^2 + (y-2)^2 = \frac{256}{13}$$

$$(4) \int \frac{3x}{x^2-2} dx = \frac{3}{2} \int \frac{2x}{x^2-1} dx$$

$$= \frac{3}{2} \log_e(x^2-1) + C$$

$$(c) \int_{\pi}^{2\pi} \cos\left(\frac{x}{3}\right) dx = \left[3 \sin \frac{x}{3} \right]_{\pi}^{2\pi}$$

$$= 3 \sin \frac{2\pi}{3} - 3 \sin \frac{\pi}{3}$$

$$= 3 \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right)$$

$$= 0$$

Question 4

$$(a) \frac{\sqrt{3}-1}{2\sqrt{3}-1} \times \frac{2\sqrt{3}+1}{2\sqrt{3}+1}$$

$$= \frac{2 \times 3 - 2\sqrt{3} + \sqrt{3} - 1}{4 \times 3 - 1}$$

$$= \frac{5 - \sqrt{3}}{11}$$

$$(b) \frac{d^2s}{dt^2} = 3t - 4$$

$$\frac{ds}{dt} = \frac{3t^2}{2} - 4t + c_1$$

$$\text{When } t=0 \frac{ds}{dt} = 0$$

$$\therefore 0 = 0 + c_1$$

$$c_1 = 0$$

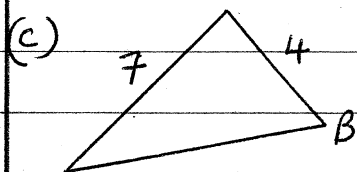
$$\therefore \frac{ds}{dt} = \frac{3t^2}{2} - 4t$$

$$\therefore s = \frac{t^3}{2} - 2t^2 + c_2$$

$$\text{When } t=0 \quad s=6$$

$$6 = 0 + c_2$$

$$\therefore s = \frac{t^3}{2} - 2t^2 + 6$$



$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{4} = \frac{2}{7}$$

$$\sin A = 4 \times \frac{2}{3} \times \frac{1}{7}$$

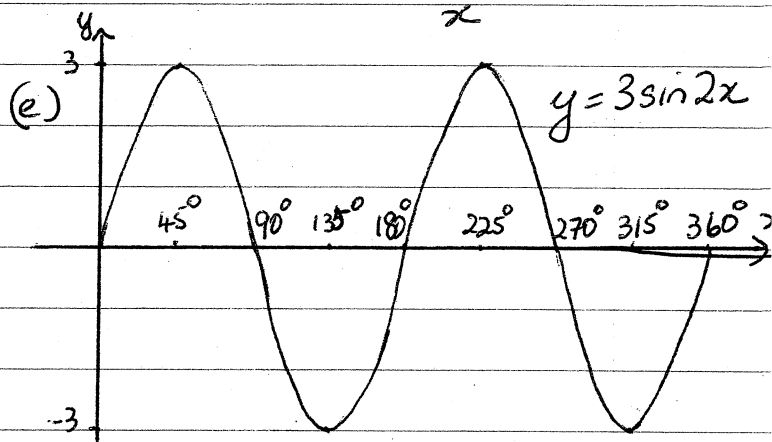
$$= \frac{8}{21}$$

$$(d) \int \frac{x^2 + x - 1}{x^2} dx$$

$$= \int 1 + \frac{1}{x} - x^{-2} dx$$

$$= x + \ln x - \frac{x^{-1}}{-1} + c$$

$$= x + \ln x + \frac{1}{x} + c$$



$$(f) 3 \times 9^x - 28 \times 3^x + 9 = 0$$

$$3 \times (3^x)^2 - 28 \times 3^x + 9 = 0$$

$$\text{Let } m = 3^x$$

$$3m^2 - 28m + 9 = 0$$

$$(3m-1)(m-9) = 0$$

$$m = \frac{1}{3}, 9$$

$$3^x = 3^{-1}, 3^2$$

$$x = -1, 2$$

Question 5

(a) $y = \frac{\tan x}{x}$

$$\frac{dy}{dx} = \frac{x \cdot \sec^2 x - \tan x \times 1}{x^2}$$

$$= \frac{x \sec^2 x - \tan x}{x^2}$$

x	0	1	2
$\frac{d^2y}{dx^2}$	-6	0	6

$\wedge$
 $\vee$

Change of concavity

$\therefore (1, -2)$ is a point of inflexion

(b) $y = x^3 - 3x^2$

(i) $\frac{dy}{dx} = 3x^2 - 6x$

$$\frac{d^2y}{dx^2} = 6x - 6$$

Stationary points occur when $\frac{dy}{dx} = 0$

i.e. $3x^2 - 6x = 0$

$$3x(x-2) = 0$$

$$x = 0, 2$$

When $x = 0$ $y = 0$

$$\frac{d^2y}{dx^2} = -6$$

$$\frac{d^2y}{dx^2} < 0$$

$\therefore (0, 0)$ is a max tp

When $x = 2$ $y = 8 - 12 = -4$

$$\frac{d^2y}{dx^2} = 6 > 0$$

$\therefore (2, -4)$ is a min tp

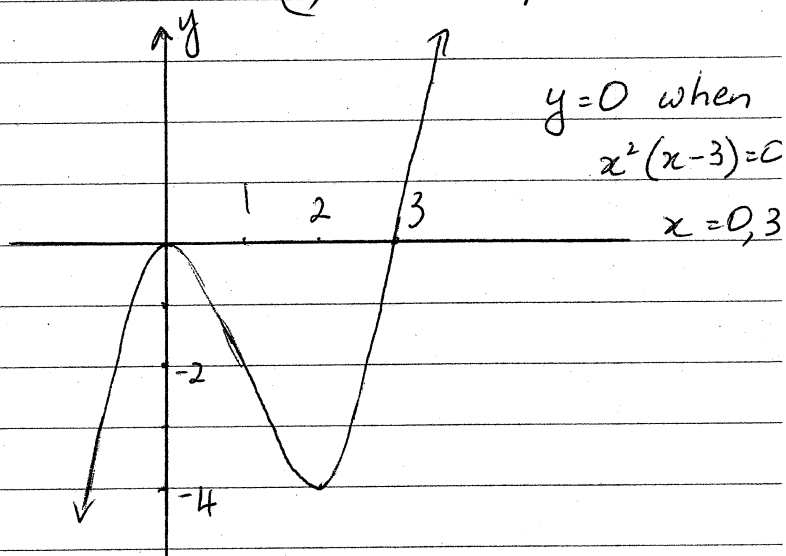
(ii) Possible inflexions when

$$\frac{d^2y}{dx^2} = 0$$

$$6x - 6 = 0$$

$$x = 1$$

$$y = -2$$



(c) $\int \sqrt[3]{x} + \sin 3x \, dx$

$$= \int x^{1/3} + \sin 3x \, dx$$

$$= \frac{x^{4/3}}{4/3} + \frac{-\cos 3x}{3} + c$$

$$= \frac{3}{4} \sqrt[3]{x^4} - \frac{\cos 3x}{3} + c$$

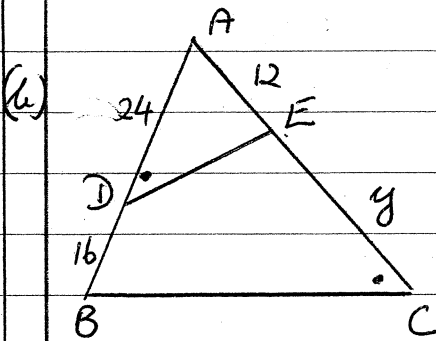
(OR $\frac{3}{4} x \sqrt[3]{x} - \frac{\cos 3x}{3} + c$)

Question 6

$$\begin{aligned}
 \text{(a) LHS} &= \frac{\sin^2 x}{1 - \cos x} + \frac{\sin^2 x}{1 + \cos x} \\
 &= \sin^2 x \left(\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} \right) \\
 &= \sin^2 x \frac{(1 + \cos x + 1 - \cos x)}{1 - \cos^2 x} \\
 &= \sin^2 x \times \frac{2}{\sin^2 x}
 \end{aligned}$$

$$= 2$$

$$= \text{RHS}$$



In Δ s AED, ABC

$\angle DAE$ is common

$\angle ADE = \angle ACB$ (given)

$\therefore \Delta ADE \parallel \Delta ACB$ (equiangular)

$\frac{AD}{AC} = \frac{AE}{AB} = \frac{DE}{CB}$ (corresponding sides in similar triangles)

$$\frac{24}{12+y} = \frac{12}{40}$$

$$\frac{2}{12+y} = \frac{1}{40}$$

$$12+y = 80$$

$$y = 68$$

$$\begin{aligned}
 \text{(c) } x^2 - 4x - 2y + 8 &= 0 \\
 x^2 - 4x &= 2y - 8 \\
 x^2 - 4x + 4 &= 2y - 8 + 4 \\
 (x-2)^2 &= 2y - 4 \\
 &= 2(y-2) \\
 &= 4 \times \frac{1}{2} (y-2)
 \end{aligned}$$

(i) Vertex is $(2, 2)$

(ii) Focal length = $\frac{1}{2}$
Focus is $(2, 2\frac{1}{2})$ or $(2, \frac{5}{2})$

$$\begin{aligned}
 \text{(iii) } 2y &= x^2 - 4x + 8 \\
 y &= \frac{1}{2}x^2 - 2x + 4 \\
 \frac{dy}{dx} &= x - 2
 \end{aligned}$$

$$\text{When } x = 0, \frac{dy}{dx} = -2$$

$\therefore$ Grad of normal at $(0, 4)$ is $\frac{1}{2}$

Question 7

(a) (i) (1) $l = r\theta$

(2) $A = \frac{1}{2}r^2\theta$

(ii) Perimeter = $2r + r\theta$

$$2r + r\theta = 12$$

$$r(\theta + 2) = 12$$

$$r = \frac{12}{\theta + 2}$$

$$\therefore A = \frac{1}{2} \times \left(\frac{12}{\theta + 2} \right)^2 \times \theta$$

$$= \frac{72\theta}{(\theta + 2)^2}$$

(iii) $\frac{dA}{d\theta} = \frac{72(\theta + 2)^2 - 72\theta \cdot 2(\theta + 2)}{(\theta + 2)^4}$

$$= \frac{72(\theta + 2)(\theta + 2 - 2\theta)}{(\theta + 2)^4}$$

$$= \frac{72(2 - \theta)}{(\theta + 2)^3}$$

Stationary points occur when $\frac{dA}{d\theta} = 0$

$$\frac{72(2 - \theta)}{(\theta + 2)^3} = 0$$

$$\theta = 2$$

θ	1	2	3
$\frac{dA}{d\theta}$	$\frac{72}{27}$	0	$-\frac{72}{125}$

+ / - \ -

$\therefore$ Max tp when $\theta = 2$

When $\theta = 2$ $A = \frac{72 \times 2}{(2 + 2)^2} = 9$

$\therefore$ Max area = 9m^2

(b) $A = \int_0^{\frac{\pi}{4}} (\sec^2 x - x) dx$

$$= \left[\tan x - \frac{x^2}{2} \right]_0^{\frac{\pi}{4}}$$

$$= \left(\tan \frac{\pi}{4} - \frac{1}{2} \cdot \frac{\pi^2}{16} \right) - (0 - 0)$$

$$= 1 - \frac{\pi^2}{32}$$

$$\text{Area} = 0.69157 \dots$$

$$= 0.69 \text{ unit}^2 \text{ (2dp)}$$

(c) $4x^2 - mx + 9 = 0$

has real roots when $\Delta \geq 0$

$$\Delta = (-m)^2 - 4 \times 4 \times 9$$

$$= m^2 - 144$$

$\therefore$ Real roots when

$$(m - 12)(m + 12) \geq 0$$

$$m \leq -12 \text{ or } m \geq 12$$

Question 8

- (a) Let a be 1st term and r the common ratio

$$t_2 = ar = \frac{3}{8} \quad (1)$$

$$t_7 = ar^6 = 12 \quad (2)$$

$$(2) \div (1) \quad \frac{ar^6}{ar} = 12 \div \frac{3}{8}$$

$$r^5 = 32$$

$$r = 2$$

$$a \times 2 = \frac{3}{8}$$

$$a = \frac{3}{16}$$

$$\begin{aligned} \therefore t_{14} &= ar^{13} \\ &= \frac{3}{16} \times 2^{13} \\ &= 3 \times 2^9 \\ &= 1536 \end{aligned}$$

(b) $3x^2 + 4x - 3 = 0$

(i) $\alpha + \beta = -\frac{4}{3}$

(ii) $\alpha\beta = -\frac{3}{3}$
 $= -1$

(iii) $2(\alpha^2 + \beta^2) = 2[(\alpha + \beta)^2 - 2\alpha\beta]$
 $= 2\left[\left(-\frac{4}{3}\right)^2 - 2 \times -1\right]$
 $= 2\left[\frac{16}{9} + 2\right]$
 $= 7\frac{5}{9} \left(= \frac{68}{9}\right)$

(c) $\frac{x^2}{2} + y^2 = 8 \quad x \geq 0$

When $y = 0$ $\frac{x^2}{2} = 8$
 $x = 4 \quad (x \geq 0)$

$$\begin{aligned} V &= \pi \int_0^4 y^2 dx \\ &= \pi \int_0^4 8 - \frac{x^2}{2} dx \\ &= \pi \left[8x - \frac{x^3}{6} \right]_0^4 \\ &= \pi \left[(32 - \frac{64}{6}) - (0 - 0) \right] \\ \text{Volume} &= \frac{64\pi}{3} \text{ units}^3 \end{aligned}$$

(d) Interest rate = 6% p.a. = 0.5% per month

(i) $V = 10000 \times 1.005^{24}$
 $= 11271.59776$

Value is \$11271.60 (nearest cent)

(ii) Interest for 2 yrs = \$1271.60

Interest for 1 yr = \$635.80

Simple Interest rate = $\frac{635.8}{10000} \times 100\%$
 $= 6.358\% \text{ p.a.}$
 $\approx 6.36\% \text{ p.a.}$

Question 9

(a)(i) $\frac{d(x \ln x - x)}{dx}$

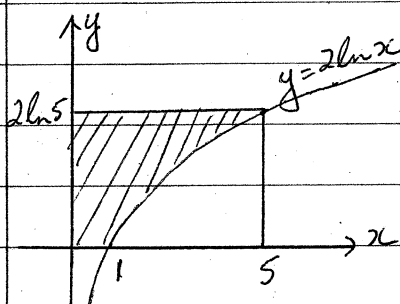
$$= 1 \cdot \ln x + x \times \frac{1}{x} - 1$$

$$= \ln x + 1 - 1$$

$$= \ln x$$

(ii) $\int \ln x^2 dx = 2 \int \ln x dx$
 $= 2(x \ln x - x) + c$
 $= 2x \ln x - 2x + c$

(iii) When $x=5$ $y = \ln 5^2$
 $= 2 \ln 5$



Shaded area

$$= \text{Area of rectangle} - \int_1^5 2 \ln x dx$$

$$= 5 \times 2 \ln 5 - 2[x \ln x - x]_1^5$$

$$= 10 \ln 5 - 2\{(5 \ln 5 - 5) - (\ln 1 - 1)\}$$

$$= 10 \ln 5 - 10 \ln 5 + 10 - 2$$

$$= 8$$

OR (Otherwise)

$$y = \ln x^2$$

$$x = e^{y/2}$$

$$x = e^{y/2}$$

$$A = \int_0^{2 \ln 5} e^{y/2} dy$$

$$= 2[e^{y/2}]_0^{2 \ln 5}$$

$$= 2(e^{\ln 5} - e^0)$$

$$= 2(5 - 1) = 8$$

(b)(i) $y = \frac{1}{x} + \log x$

Domain: $x > 0$

(ii) $y = x^{-1} + \log x$

$$\frac{dy}{dx} = -1x^{-2} + \frac{1}{x}$$

$$= -\frac{1}{x^2} + \frac{1}{x}$$

$$= \frac{-1+x}{x^2} = \frac{x-1}{x^2}$$

$$\frac{d^2y}{dx^2} = 2x^{-3} - \frac{1}{x^2}$$

$$= \frac{2}{x^3} - \frac{1}{x^2}$$

$$= \frac{2-x}{x^3}$$

(iii) Stationary points occur when $\frac{dy}{dx} = 0$

$$\frac{x-1}{x^2} = 0$$

$$x = 1$$

$$y = \frac{1}{1} + \log 1$$

$$= 1$$

When $x=1$ $\frac{d^2y}{dx^2} = \frac{2-1}{1^3} = 1 > 0$

$\therefore (1, 1)$ is a maximum tp.

(c) $\int_1^k \frac{4}{x^2} dx = \int_1^k 4x^{-2} dx$

$$= \left[4x^{-1} \right]_1^k$$

$$= \left[-\frac{4}{x} \right]_1^k$$

$$\therefore 3 = -\frac{4}{k} - (-4)$$

$$= 4 - \frac{4}{k}$$

$$\frac{4}{k} = 1$$

$$k = 4$$

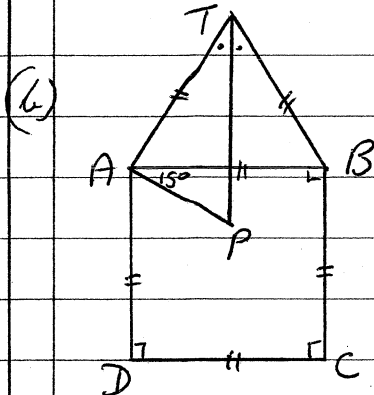
Question 10

(a)(i) $x = 40$

(Use exterior $\angle$ of triangles,
angle sum of Δ)

(ii) $x = 22.5$

(Use equal $\angle$ s in isosceles
 Δ and exterior $\angle$ of Δ)



(i) $\angle TAB = 60^\circ$ ($\angle$ in equilateral Δ)

$\angle PAT = 60^\circ + 15^\circ = 75^\circ$

(ii) $\angle BAD = 90^\circ$ (angle in a square)

$\therefore \angle DAP = 90^\circ - 15^\circ$

$= 75^\circ$

In Δ s TAP, DAP

$AT = AB$ (equal sides in equilateral Δ)

$AD = AB$ (equal sides of square)

$\therefore AT = AD (= AB)$

$\angle TAP = \angle DAP (= 75^\circ)$

AP is common

$\therefore \Delta TAP \equiv \Delta DAP$ (SAS)

(ii) $A_6 = 20000 - 6m$

$A_7 = A_6 + \text{Interest} - m$

$= A_6 \times 1.01 - m$

$= (20000 - 6m) \times 1.01 - m$

$A_8 = A_7 \times 1.01 - m$

$= (20000 - 6m) \times 1.01^2 - m \times 1.01 - m$

$= (20000 - 6m) \times 1.01^2 - m(1 + 1.01)$

By the same pattern

$A_n = (20000 - 6m) \times 1.01^{n-6} - m(1 + 1.01 + \dots + 1.01^{n-6})$

$\therefore A_{60} = (20000 - 6m) \times 1.01^{54} - m(1 + 1.01 + \dots + 1.01^5)$

$= (20000 - 6m) \times 1.01^{54} - m \frac{(1.01^{54} - 1)}{1.01 - 1}$

$= (20000 - 6m) \times 1.01^{54} - \frac{m(1.01^{54} - 1)}{0.01}$

Since loan is repaid after 5yrs

$A_{60} = 0$

$\therefore 0 = 20000 \times 1.01^{54} - 6m \times 1.01^{54} - 100m(1.01^{54} - 1)$

$6m \times 1.01^{54} + 100m(1.01^{54} - 1) = 20000 \times 1.01^{54}$

$m[6 \times 1.01^{54} + 100 \times 1.01^{54} - 100] = 20000 \times 1.01^{54}$

$m = \frac{20000 \times 1.01^{54}}{106 \times 1.01^{54} - 100}$

$= \frac{20000 \times 1.01^{54}}{106 \times 1.01^{54} - 100}$

$= 420.4448 \dots$

(c) (i) $A_1 = 20000 - m$

$A_2 = 20000 - m - m$

$= 20000 - 2m$

$A_4 = 20000 - 4m$

$m = 420.44$ (2 dp)